

3D Scene Reconstruction of Point Cloud Data: A Lightweight Procedural Approach

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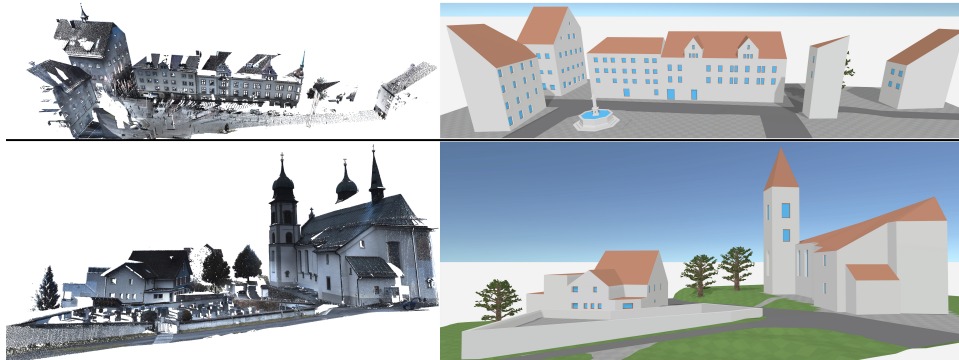


Figure 1: Two scenes from the Semantic3d.net dataset, with the point cloud on the left and reconstruction on the right.

Abstract

In this paper, we present an approach that takes point cloud data of a scene and automatically produces a lightweight 3D reconstruction of the buildings and their facade details found in the scene. Our 3D reconstructions are described using a procedural grammar to allow for light-weight storage and processing, as well as scalability. The preliminary evaluation of our approach considers the terrestrial laser scan dataset semantic3d.net that includes a wide range of outdoor scenes and feature considerably variety in terms of the density of the point clouds. Our results show that our approach is able to extract most of the buildings and correctly detect most of the facade openings.

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1 Introduction

In this paper, we propose an end-to-end approach that, given a point cloud scene, automatically produces lightweight 3D reconstructions of the buildings and their facade details found in the scene. The 3D reconstructions are described using a procedural grammar to allow

for light-weight storage and processing, as well as scalability. Specifically, we consider the shape grammar introduced in [Bohlender et al. 2023]. We present a preliminary evaluation of our approach, which considers the scenes from the semantic3d.net [Hackel et al. 2017] dataset (including a wide range of outdoor scenes and variety in terms of the density of the point clouds). On all scenes, we found that our approach reconstructs most of the buildings and most of the details in their facades. In Figure 1, we show two scenes from the Semantic3D dataset and the corresponding reconstructions¹.

2 Related Work

The existing literature for scene reconstruction from point clouds focuses on aerial laser scan data (e.g., [Zhang et al. 2018]). However, our approach is based on ground-based laser scans, as it aims to reconstruct facade details in addition to building masses. Approaches to building reconstruction from singled out building point clouds consist of semi-automated approaches [Nan et al. 2010], plane detection based approaches (e.g., [Nan and Wonka 2017]), and line segment based approaches (e.g., [Kelly et al. 2017]). We combine the underlying ideas of the plane detection and the line segment linking approaches into our footprint detection method. For facade reconstruction on point cloud data, [Mesolongitis and Stamos 2012] only detects windows, while [Li et al. 2017] and [Xiong et al. 2013] also perform reconstruction. Our approach combines the ideas of turning facades into depth maps of [Li et al. 2017] and windows as areas behind walls of [Mesolongitis and Stamos 2012] and proposes a novel approach based on edge detection on the depth map and histogram analysis of areas of interest. For the grammar induction, we take inspiration from approaches with pre-labeled terminals (e.g., [Weissenberg et al. 2013]) and recursively create a parse tree.

¹Elements that are not part of our reconstruction approach, such as trees and roofs, were added using Unity.

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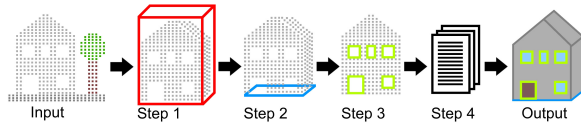


Figure 2: Overview of our reconstruction approach.

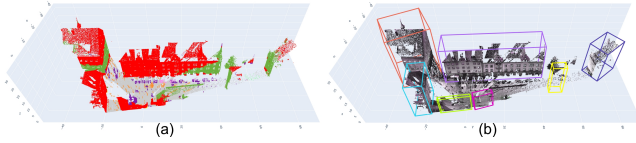


Figure 3: Semantic segmentation by RandLA-Net (a). Buildings extracted by our approach (b).

Table 1: Findings across the scenes of semantic3D.net.

Type	Total	Found	Discarded	Misidentified
Buildings	211	70.6%	13.7%	6.0%
Facade Openings	1,681	78.6%	–	12.7%

3 Our Approach

Figure 2 shows an overview of our approach. As an input we take a point cloud of a scene. In step 1, we semantically segment the point cloud, extract the individual clusters of building points and output the bounding box around each cluster. In step 2, we use the bounding box and the semantic segmentation to extract the points related to the building from the scene, detect the walls of the building to form the footprint from the line segments extracted from each wall. In step 3, we use the footprint to extract the points related to a facade, create a depth map from the points, and detect the facade openings from the depth map, that are regularized and used to search for missed facade openings along soft grids. In step 4, we induce a procedural shape grammar description from the detected footprint and facade openings with templates for the building mass and a recursive algorithm to create the splitting rule for each facade.

4 Evaluation

As shown in Figure 3, we build on a semantic segmentation given by the RandLA-Net [Hu et al. 2020] (a) to extract the buildings, each marked by a bounding box (b).

In Table 1 we show the quantitative results of our evaluation on all 19 real world scenes of the semantic3d.net dataset, – marks in-applicability. The total refers to the number of type instances across all 19 scenes. We correctly find 70.6% of the buildings and find but discard another 13.7%. 6.0% of extractions are other objects, e.g., garages. We correctly reconstruct 78.6% of all facade openings. 12.7% of all reconstructed facade openings stem from occlusions.

Figure 4 shows examples of reconstructed buildings. Blue outlines mark building footprints more complex than rectangles. Solid red circles mark correctly reconstructed facade openings near occlusions. Solid green circles mark correctly reconstructed, small facade openings. Differently colored dots mark facade openings on the same soft grid. Solid orange circles mark positions on the soft grids

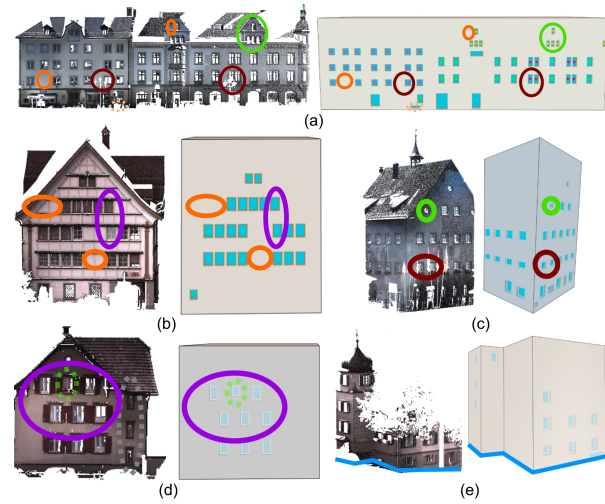


Figure 4: Examples of reconstructed buildings.

with correctly not inserted facade openings. Solid purple circles mark correctly not enforced grids. Occlusions can introduce (dotted orange circles) or deform (dotted green circles) facade openings. Deformation can also be due to blocking items, e.g. flowerpots.

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